House Price Regression - Submission Document

Task: House Price Prediction Regression | Student: Jeswanth Reddy Reddem

Submission Links

Replace these placeholders with real GitHub / Google Drive links before final upload.

Artifact	Link
GitHub Repository	GitHub repo link
Notebook Link	GitHub notebook link
Saved Model Link	GitHub model file link
Screenshots / Plots Link	GitHub plots folder link

1-Page Executive Summary

This project builds a regression model to predict house prices using structured housing attributes such as bedrooms, bathrooms, living area, lot size, floors, waterfront status, view, condition, year built, renovation year, city, and zipcode.

The workflow includes exploratory data analysis, target cleaning, log transformation of skewed variables, categorical encoding, numerical scaling, missing-value handling, model comparison, and residual analysis. Rows with non-positive prices were removed before log-target regression.

Feature engineering creates sale year, sale month, sale day of week, house age, renovation indicator, renovation age, and extracted zipcode. Street is removed because it has very high cardinality and can overfit; city and zipcode retain useful location information.

Three models are compared: Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor. The selected best model is Linear Regression, based on the lowest RMSE on the test set.

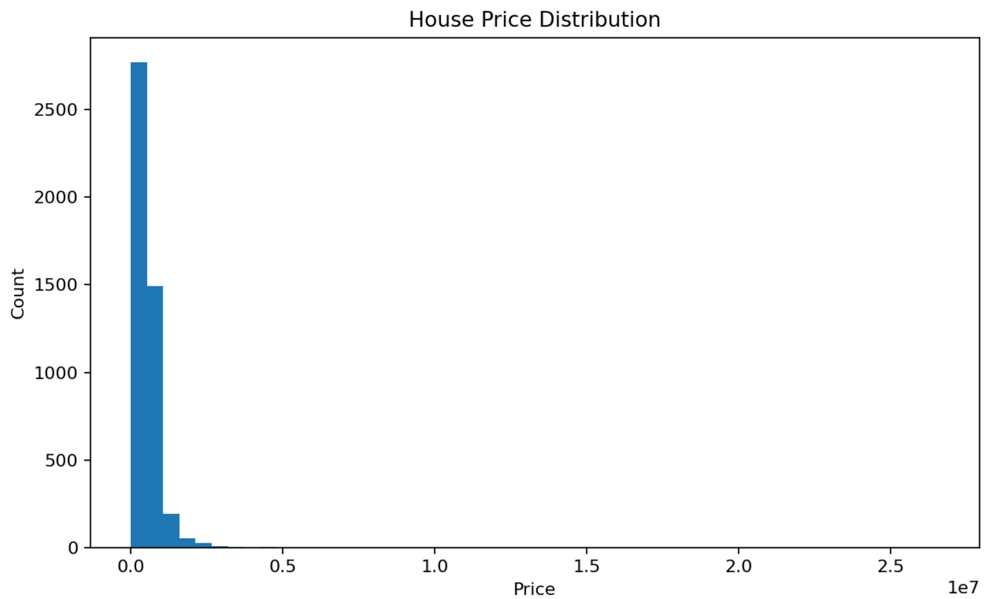
The final model is saved in JOBLIB and Pickle formats. The notebook and inference script show how to load the model and predict the price for a new house record.

Model Evaluation Summary

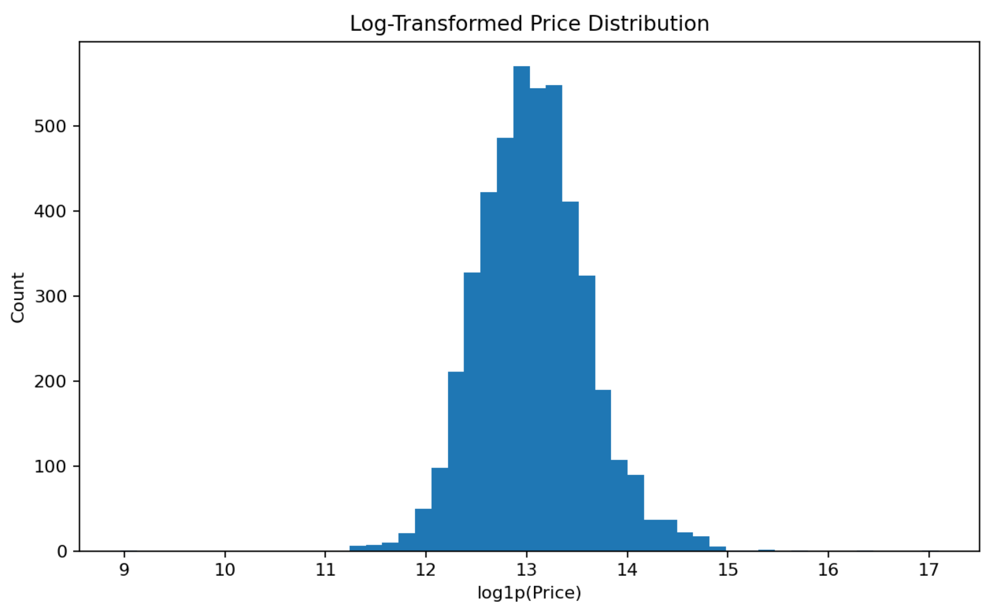
Model	RMSE	MAE	R2
Linear Regression	173,370.22	87,261.18	0.7980
Random Forest	220,447.47	111,007.93	0.6733
Gradient Boosting	239,498.03	124,602.91	0.6144

Screenshots / Plots

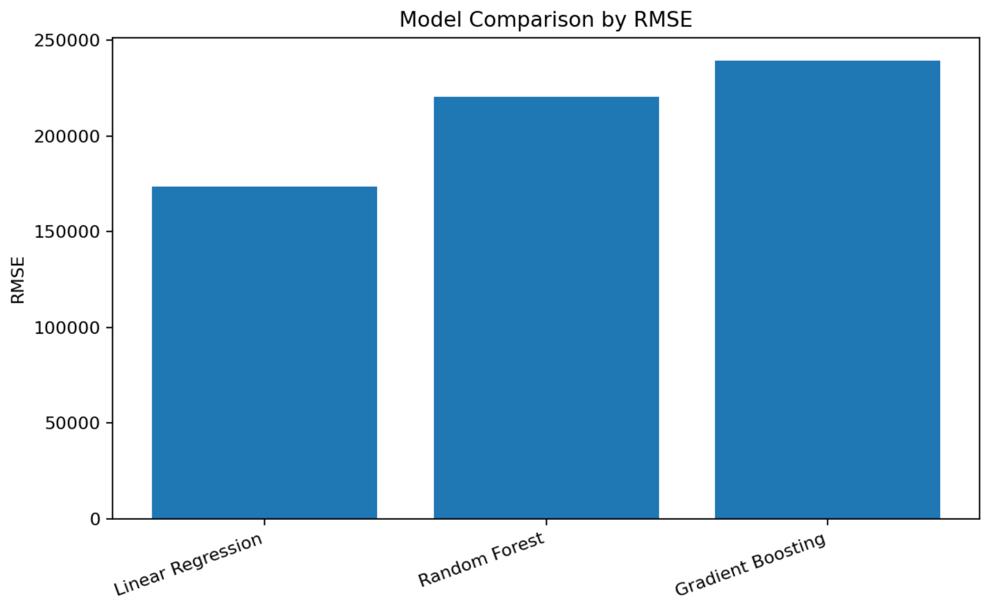
Raw price distribution



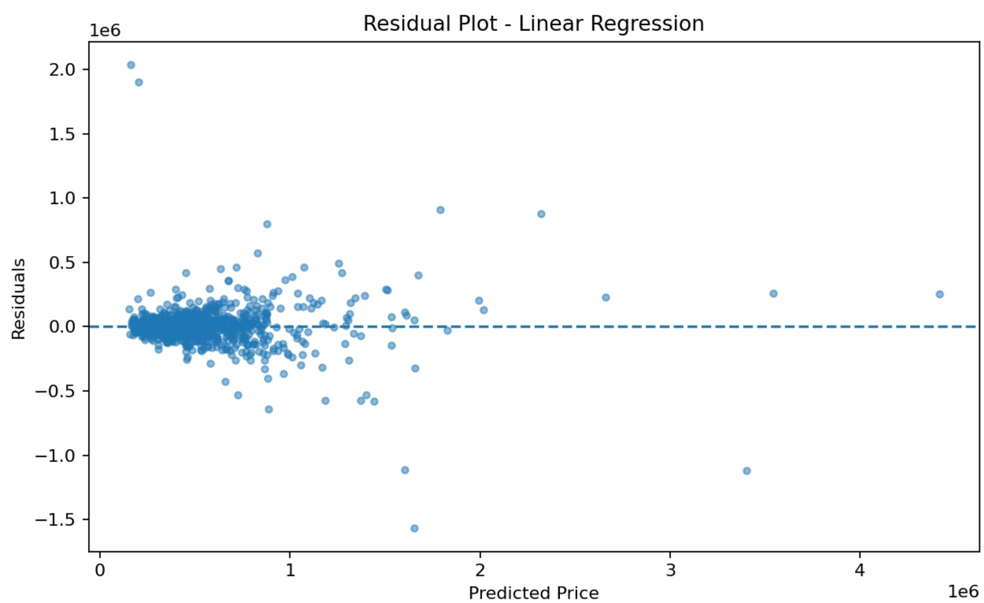
Log-transformed price distribution



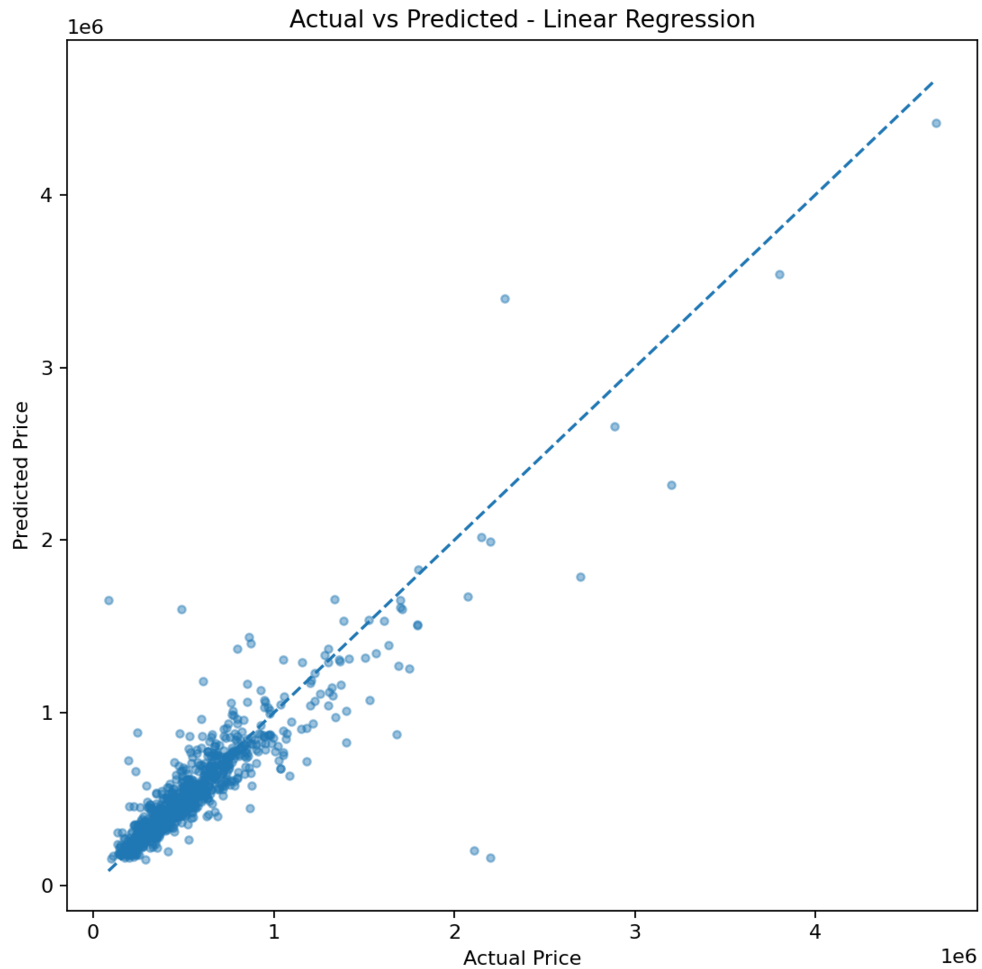
Model comparison by RMSE



Residual plot



Actual vs predicted price



Commands to Reproduce

```
pip install -r requirements.txt

python src/train_model.py

python src/inference.py
```

Submission Checklist

- Push notebooks and code to GitHub.
- Upload large files/model to Google Drive if required.
- Make all links viewable.
- Paste links on the Task Submission page and mention task number clearly.